SUYASH PRAKASH

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LinkedIn link

EDUCATION

BTech - Robotics and Autonomous systems	Plaksha University, Mohali	7.97	2024 - Present
12th - CBSE	Saint Soldier International school, Chandigarh	91.8	2024
10th - CBSE	Saint Soldier International school, Chandigarh	91	2022

PROJECTS

Consultant and FPV drone frame designer | Self Oct 2022 - Oct 2025

- Self-learned **professional FPV drone pilot** and designer with hands-on experience building **custom racing and freestyle drones**, mentoring pilots in **PID tuning**, and **analyzing blackbox data** to improve mechanical resonance and frame rigidity.
- Invented and developed a **patented** carbon-fiber FPV drone frame, designed in CAD (FreeCAD), fabricated in T300 carbon fiber in China, and successfully commercialized with **15 units sold** in the Indian market..

Mechanical Lead for CANSAT 2025 Competition| Robotics Club - Prof Shashank Tamaskar Oct 2024 - Feb 2025

- Competition required building a can-sized satellite that descends using autogyro blades while gathering flight data.
- Designed the complete assembly of satellite body, internal structure and autogyro blades using fusion360 CAD.
- Achieved a **PDR result of 92.4%.**

Team Lead / CTO for URC 2026 Competition | Robotics Club - Prof Deepan Muthirayan March 2025 - February 2026

- Competition requires building a mars rover prototype which can **autonomously traverse** dessert terrain equipped with **long range communication** systems, bio-sciences module for **life-detection** and a **6-DOF manipulator** to perform certain tasks.
- Did **system designing** for the entire project along with creating **timelines** and establishing **workflows**.
- Facilitated the **integration** of various **sub-systems** in our early prototype.
- Designed the **RF communication** module and the **low level control algorithm** for the rover.
- **Cleared the PDR round for the 2026 Season.**

Bi-copter Build | Self June 2026 - July 2026

- Designed and Built a custom bi-copter with two drone motors and servo actuation for thrust vectoring.
- Built it using carbon tubes and rods. Custom designed the frame in CAD
- Used INAV software to configure it and PID tuned it to fly stable.

Data-Driven Feedforward Motor Model Development| B-Tech Project - Prof Andy Ruina Jan 2026 - Present

- Developing a data-driven **Feedforward** model of an DC motor using **experimentally acquired** electro-mechanical system data.
- Designing instrumentation to acquire synchronized **torque, speed, current, voltage, temperature**, and **gearbox backlash** measurements for **deadband analysis**.
- Investigating and comparing **data-driven motor modeling** techniques—including **lookup tables, neural networks, and physics-informed machine learning**—to generate accurate inverse motor models for controls.

POSITIONS OF RESPONSIBILITY

CTO | Kalki Robotics Club - Plaksha University March 2025 - Present

- Over-seeing the technical aspect of both URC and CANSAT competitions.
- Facilitating fabrication of the designs by coordinating with our labs, makerspace and also external vendors.
- Teach new-comers / juniors about mechatronics.

Robotics Lab volunteer | Plaksha University August 2025 - present

- Present the projects going on in the lab such as URC , CANSAT, and other course project to founders, guests and new comers.

ACHIEVEMENTS

- **Intellectual Property (IP)** Quadcopter Frame Design registered with The Patent Office in India Design No. 392685-001
- AP calculus BC - 4/5 - 2024.
- Achieved 92.4% PDR score in CANSAT 2025.
- Cleared PDR round for URC 2026.

SKILLS

- **FPV** Freestyle and Cinematic piloting (YouTube)| drone assembly / prototyping
- **3D-CAD Design / Rapid Prototyping** : Fusion360, FreeCAD, Blender, Cura, Prusa Slicer, RDWorks
- Operating 3D printers, laser machines and basic power / hand tools for wood working.
- **Electronics IOT :** SMD soldering, circuit prototyping, Arduino C.
- Proficient with DC and BLDC motors along with micro-controller (Ardunio, teensy, raspberry pi) and signal types (I2C ,SPI etc)
- **Programming Languages:** Python (Numpy ,Sympy, Scipy, matplotlib, pandas) , C++ , Matlab
- **Leadership Skills :** Project Management, System Design, timeline creations